

Functions



Evaluating functions

- 1 If $f(x) = 3x - 4$, find $f(5)$.
 - 2 If $f(x) = x^2 - 2x + 1$, find $f(-2)$.
 - 3 If $g(x) = \sqrt{x + 5}$, find $g(4)$.
 - 4 If $h(x) = \frac{2x}{x - 1}$, find $h(3)$.
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Domain and range

- 5 State the domain of $f(x) = \sqrt{x - 3}$.
 - 6 State the domain of $f(x) = \frac{1}{x - 5}$.
 - 7 State the domain of $f(x) = \sqrt{7 - x}$.
 - 8 State the range of $f(x) = x^2 + 2$.
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Transformations: shifts

- 9 The graph of $y = x^2$ is shifted 3 units right and 2 units down. Write the new equation.
 - 10 The graph of $y = |x|$ is shifted 4 units left and 1 unit up. Write the new equation.
 - 11 Describe the transformation that takes $y = x^2$ to $y = (x + 2)^2 - 5$.
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Transformations: reflections and stretches

- 12 The graph of $y = x^2$ is reflected over the x -axis and stretched vertically by a factor of 3. Write the new equation.
- 13 Describe the transformation that takes $y = \sqrt{x}$ to $y = 2\sqrt{x}$.
- 14 Describe the transformation that takes $y = x^2$ to $y = -(x - 1)^2 + 4$.

Composite functions

- 15 Let $f(x) = 2x + 1$ and $g(x) = x^2 - 3$. Find $(f \circ g)(2)$.
- 16 Using the same f and g , find $(g \circ f)(2)$.
- 17 Using the same f and g , find a general formula for $(f \circ g)(x)$.
- 18 Using the same f and g , find a general formula for $(g \circ f)(x)$.
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Inverse functions

- 19 Find the inverse of $f(x) = 3x - 2$.
- 20 Find the inverse of $f(x) = \frac{x - 4}{5}$.
- 21 Show that $f(x) = 2x + 6$ and $g(x) = \frac{x - 6}{2}$ are inverses by computing $f(g(x))$.
- 22 Find the inverse of $f(x) = x^3 + 1$.
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Piecewise functions

- 23 A function is defined by $f(x) = x + 2$ for $x < 1$, and $f(x) = 3x - 1$ for $x \geq 1$. Find $f(-2)$.
- 24 Using the same piecewise function, find $f(1)$.
- 25 Using the same piecewise function, find $f(4)$.
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Even and odd functions

- 26 Determine whether $f(x) = x^2 - 3$ is even, odd, or neither.
- 27 Determine whether $f(x) = x^3 - x$ is even, odd, or neither.
- 28 Determine whether $f(x) = x^2 + x$ is even, odd, or neither.
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Increasing, decreasing, and average rate of change

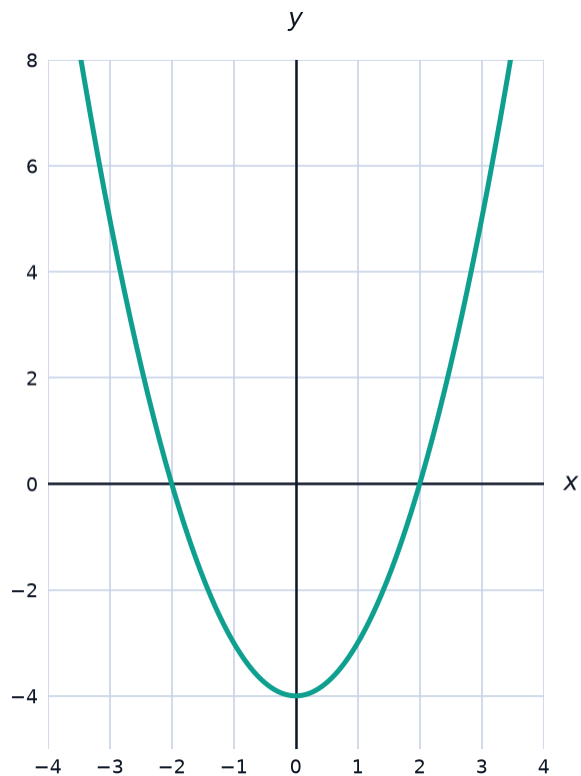
- 29 Find the average rate of change of $f(x) = x^2$ from $x = 1$ to $x = 4$.
- 30 Find the average rate of change of $f(x) = 2x^2 - 3x$ from $x = 0$ to $x = 3$.
- 31 State the interval on which $f(x) = (x - 2)^2$ is decreasing.
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Operations on functions

- 32 Let $f(x) = x + 3$ and $g(x) = x^2 - 1$. Find $(f + g)(x)$.
- 33 Using the same f and g , find $(f \cdot g)(2)$.
- 34 Using the same f and g , find $(f/g)(3)$.
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Reading a function graph

- 35 The graph of $y = f(x) = x^2 - 4$ is shown.



- a State the zeros of f .
- b State the domain and range of f .

Word problems: functions in context

- 36** This question has two parts. A company's daily profit (in dollars) from selling x units is modeled by $P(x) = -2x^2 + 80x - 150$.
- a** Evaluate $P(10)$ to find the profit from selling 10 units.
 - b** Find the number of units x that maximizes profit (using the vertex of the parabola), and state the maximum profit.
- 37** This question has two parts. The Fahrenheit conversion of a Celsius temperature is $F(C) = \frac{9}{5}C + 32$. Meanwhile, the temperature in a city t hours after 6 a.m. is modeled by $C(t) = -0.5t^2 + 6t + 10$ (in degrees Celsius), for $0 \leq t \leq 12$.
- a** Find $C(4)$, the Celsius temperature 4 hours after 6 a.m.
 - b** Find $(F \circ C)(4)$, the Fahrenheit temperature at that time.